

Concept Review

Ultrasonic Sensors

Why Do We Need Ultrasonic Sensors?

Many mechatronics systems rely on distance sensors for obstacle avoidance, object detection, and localization. Ultrasonic sensors are included within a category of sensors that measure distance. Compared to other distance sensors that work by detecting light, ultrasonic sensors use reflected soundwaves, making them reliable across lighting conditions and for detecting objects with differing light reflectivity.

Ultrasonic Sensors

Ultrasonic sensors operate using signal emission and signal detection to perform distance estimation through the time-of-flight principle. Ultrasonic waves are sound waves above the upper limit of the human hearing range of 20 kilohertz. Ultrasonic sensors function by emitting ultrasonic waves, which are then reflected by nearby objects with a time delay proportional to the distance of the object. By measuring the amount of time taken for the emitted signal to be detected back at the sensor (the time-of-flight), the distance of objects from the sensor can be estimated.

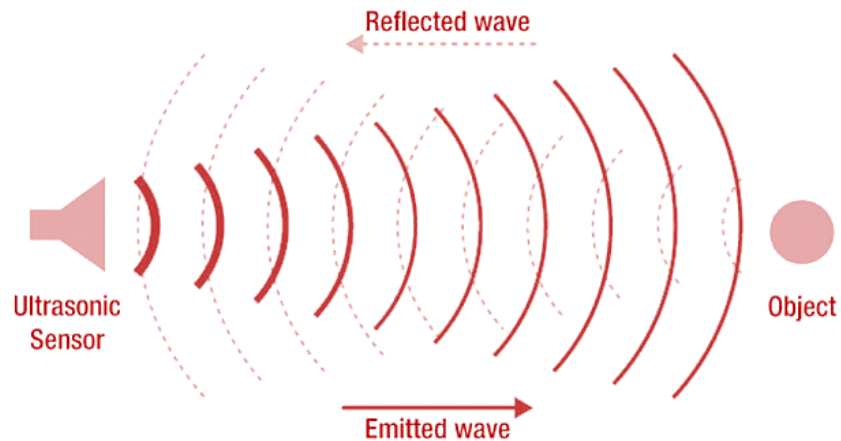


Figure 1: Working principle of an ultrasonic sensor

An ultrasonic sensor typically consists of a transmitter to produce ultrasound and a receiver to convert ultrasound to electrical signals. Another type of ultrasonic sensor uses transceivers, which performs both functions of transmitting and receiving ultrasound. Ultrasonic sensors can come in varying sizes and configurations, and two different ones are shown in Figure 2.



a. HC-SR04

b. CH201-00ABR

Figure 2: Different ultrasonic sensor configurations

Selecting an Ultrasonic Sensor

Datasheets are used to select an appropriate sensor for a specific application. For ultrasonic sensors, some key datasheet considerations include the following:

Sensing range: This is the range of distances from the sensor where objects can be detected by the sensor.

Field of view: This describes the dimensions of the area within which objects are detectable by the sensor. In sensors containing a separate emitter and receiver, the field of view is determined by the field of view of both components.

Resolution: This refers to the smallest detectable change in distance. Lower resolutions may be sufficient for less critical tasks, while high resolutions are useful for applications that require more detailed measurement, such as imaging.

Sampling rate: This value indicates how quickly the sensor can output measurements. The minimum sampling rate depends on the response time requirements for the specific application.

Comparison to Other Distance Sensors

The ultrasonic sensor is one of several types of sensors that measure distance using different technologies. The type of sensor that is best suited for an application depends on specific requirements such as cost, range, resolution, accuracy, and speed. The ultrasonic sensor is a cost-effective option for simple object detection. For applications requiring faster readings at a high accuracy, time-of-flight sensors are a more suitable choice but can come at a higher cost. When choosing between a light-based or a sound-based distance sensor, additional considerations include both the target object and operating environment. The performance of ultrasonic sensors is unaffected by certain material properties of the target object such as reflectivity and color, or by environmental factors such as light, dust, and humidity, unlike light-based sensors.

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